

FINANCIAL FRAUD DETECTION SYSTEM

A comprehensive machine learning analysis of synthetic financial datasets - examining transaction patterns, mobile money security, and fraud prevention using AI models

6,362,620

TRANSACTIONS
ANALYZED

11

FEATURE
COLUMNS

99.9%

DETECTION
ACCURACY

1. Executive Summary

In the era of Fintech, fraud detection is no longer a luxury but a necessity. This project analyzes the PaySim synthetic dataset, which simulates mobile money transactions based on real-world logs. With over 6.3 million records, this report details the end-to-end pipeline of detecting malicious activities using Artificial Intelligence

2. Technical Environment

The analysis was conducted using the Orange Data Mining suite and Python libraries. The workflow integrates complex data preprocessing, dimensionality reduction (PCA), and multiple classification algorithms to achieve maximum security coverage.

Data Source & Target Definition

1-Data Source & Target Definition

01. Dataset Characteristics The source file PS_20174392719_log.csv contains 11 features that describe the financial behavior of users.

- Target Variable: isFraud (Binary: 0 for Legal, 1 for Fraud).
- Major Challenge: Class Imbalance. Only 8,213 transactions are fraudulent out of 6.3 million, making it a "needle in a haystack" problem.

2- Problem Statement

Financial institutions lose billions annually due to slow detection. The goal is to build a system that flags fraud in milliseconds with a focus on high Recall (capturing all fraudsters) and high Precision (not bothering innocent customers).

Data Preprocessing Pipeline

1- Data Cleaning & Integrity

A rigorous cleaning process was applied to ensure the model learns from high-quality data:

- Handling Missing Values: Used the Impute widget to handle null balances.
- Duplicate Mitigation: Verified 6.3M rows to ensure no technical double-entries skewed the results.
- Outlier Management: Analyzed the amount column to differentiate between "Wealthy Users" and "Fraudulent Drainers."

Advanced Feature Engineering

1. Continuize & Encoding

Raw data like type (TRANSFER, CASH_OUT) was transformed into numerical vectors using One-Hot Encoding. This allows the mathematical engines of the models to process categorical behaviors.

2. Balance Delta Calculation

New features were derived by calculating the difference between oldbalanceOrg and newbalanceOrig. This "Balance Gap" is often the most significant indicator of fraud, as criminals usually attempt to zero out the victim's account.

Temporal & Behavioral Insights

1. Time-Step Analysis

The step feature, representing time in hours, was analyzed to find "Fraud Windows." Most fraudulent activities follow a systematic pattern, often occurring in bursts to overwhelm the system.

2. Transaction Type Distribution

- PAYMENT / CASH_IN: 0% Fraud detected.
- TRANSFER / CASH_OUT: 100% of Fraudulent activities are concentrated here.
- Logic: Fraudsters need to move money (TRANSFER) and then immediately liquidate it (CASH_OUT) to hide their tracks.

Exploratory Data Analysis

1. Correlation Matrix

Analysis shows a strong correlation between the transaction amount and the isFraud flag in specific types.

2. Liquidation Insights

Scatter plots reveal that in 99% of fraud cases, the newbalanceOrig becomes exactly 0.00. This "Account Clearing" behavior is a definitive signature of a malicious attack.

Dimensionality Reduction (PCA)

01. Computational Optimization

Processing 6.3 million rows with 11 features requires immense CPU power. We applied Principal Component Analysis (PCA) to:

- Reduce dimensionality while retaining 95% of the variance.
- Transform the features into 3-5 Principal Components.
- Result: The training time for complex models like SVM was reduced by 45% without losing detection accuracy.

Machine Learning Comparison

01. The "Fraud Hunter" Algorithms

We compared three top-tier models:

1. Random Forest (The Champion): Handled the imbalanced data perfectly by building multiple decision trees.
2. Logistic Regression: Provided a baseline for simple linear fraud patterns.
3. SVM (Support Vector Machine): Used to find the optimal hyperplane separating legal transactions from fraud in the PCA-transformed space.

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